



Investigation on the weld bead profile transformation with the keyhole and molten pool dynamic behavior simulation in high power laser welding



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ABSTRACT

Weld bead profile is highly related to the dynamic behavior of keyhole and molten pool. In this paper, a three-dimensional transient model is developed to investigate the weld bead profile transformation considering the keyhole and molten pool dynamic behaviors in high power laser welding. In the model, multiphase including the solid phase, liquid metal phase, metal vapor phase and air phase is considered. A variety of physical factors are taken into accounts such as melting and solidification, evaporation and condensation, fluid flow, Marangoni shear stress, recoil pressure and surface tension. An adaptive hybrid heat source model is used to fit well with evolution of the keyhole. Volume of fluid (VOF) method is adopted to track the multiphase interface. The weld bead profiles can be classified into three types and the welding process parameters are closely related to the weld bead profile transformation. The simulation results show that the fluid flow, temperature field and mechanical equilibrium in the molten pool are the essential factors that affect the weld bead profile transformation. The relationship among the welding process parameters, mechanical equilibrium, fluid flow and weld bead profile transformation is analyzed. The proposed numerical model exhibits good agreements with the experimental results with respect to the weld geometry.

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1. Introduction

Recently, the availability of high power laser welding paves the way for automotive manufacturing, aeronautics and astronautics, reactor vessels, and pipelines. In the high power laser welding process, the keyhole and molten pool dynamic behaviors play an important role in the welding process stability, the bead formation, and the joint quality [1,2]. The keyhole evolves frequently and the molten pool presents a complex and violent movement as a result of mechanical equilibrium [3]. The dynamic behaviors of the keyhole and the molten pool are closely related to welding process parameters, which will also affect the profile of the weld bead and joint quality. Hence, it's essential to study the relationship among weld bead profile, keyhole and molten pool dynamic behavior and welding process parameters.

A variety of numerical simulation of laser welding has been done to study the weld bead and dynamic behavior of the keyhole

and molten pool. Weld bead profiles can be simulated without considering the keyhole dynamic behavior for simplification. For example, Shanmugam et al. [4] investigated weld profiles under different welding process parameters using a 3D conical Gaussian heat source in the T-joint laser welding process. Wang et al. [5] calculated the weld bead profiles in dual-beam laser welding using a Gaussian volumetric heat source with the maximum heat flux decreasing linearly along the depth of penetration. However, ignoring the keyhole behavior may be far away from the practice. Thus recently, many researchers focus on the coupling of keyhole, molten pool and weld profile. Fabbro and Amara [6–9] have done considerable research to study the keyhole formation, its dynamics, and its relation with the molten pool movements in the laser welding. Cho et al. [10] investigated the fluid flow and keyhole collapse in high power laser welding. Temperature profile characteristics of the weld pool were analyzed to estimate the fusion zone shape. Tan et al. [11] adapted the evaporation and condensation model to analyze heat exchange and mass transfer in the keyhole and molten pool, and found that the inverse bremsstrahlung absorption of the plasma had a great effect on the temperature field

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and fluid flow of the molten pool. Courtois et al. [12] established a finite element multiphase model of solid, molten pool and metal vapor to study the keyhole oscillation as well as the porosity in the molten pool. Ronda and Siwek [13] investigated keyhole evolution and movement of metal vapor considering the effect of surface tension and recoil pressure using CFD software. Lu et al. [14] studied the relationship between the porosity of the molten pool and the dynamic behavior of the molten pool in different process parameters. The model involved the bubble movement and the melting and solidification. It was found that the number of the porosity was affected by the keyhole collapse frequency. Zhang et al. [15] undertook the simulation of full penetration laser welding of thick steel plate with high power laser. VOF (volume of fluid) method coupled with ray-tracing algorithm was adopted in the model to simulate the keyhole and molten pool dynamics and the weld bead profiles. Pang et al. [16] studied the coupling of three-dimensional heat transfer, fluid flow and keyhole evolutions in the welding process. The formation process of a collapsing keyhole with bubble under certain conditions was obtained. A conclusion was reached that the keyhole instability was closely related to the weld pool dynamics. Sohail et al. [17] aimed to study the influence of gravity on the penetration depth under four different angular positions. Streamlines of the molten pool were clearly displayed to illustrate the fluid flow, how the keyhole collapsed and the bubble moved during the welding process.

From the literature mentioned above, most of the previous studies focused on the dynamic behavior of the keyhole and molten pool, and the influence of process parameters on the weld bead profile. However, seldom research has been done about the classification of weld bead profiles and analysis of the distribution characteristics of process parameters in different types of weld bead profiles. Correspondingly, investigations on the effect of the keyhole and molten pool dynamic behavior on the weld bead profile are also rare. Thus, this paper will investigate reasons for the weld bead profile transformation to have a better understanding of the relationship among welding process parameters, dynamics of keyhole and molten pool and weld bead profile.

The rest of the paper is summarized as follows: In Section 2, the model is established including governing equations, laser heat source, evaporation and condensation, keyhole tracking model and boundary conditions. Experiments are conducted in Section 3. Results and discussions consisting of experiment results and weld pool simulation are described in Section 4 and finally, the conclusions are obtained.

2. Numerical modeling

2.1. Governing equations

The welding process consists of a variety of complex physical phenomena such as fluid flow, heat transfer, mass transfer and phase interaction. To simplify the simulation model, several assumptions are made: (1) The molten metal in weld pool is laminar, incompressible and Newtonian fluid; (2) The material is isotropic homogeneous and its properties are functions of temperature only; (3) No permeation is considered among the phases. (4) Plasma in the welding process is ignored. Based on these assumptions, the governing equations for continuity, momentum, and energy equation are respectively described as follows:

Continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) + S_m = 0 \quad (1)$$

where ρ is the fluid density, $\vec{V}$ is the velocity vector, and S_m is mass source term.

Momentum equation

$$\frac{\partial \rho \vec{V}}{\partial t} + \nabla \cdot (\rho \vec{V} \vec{V}) = -\nabla P + \nabla \cdot (\mu \nabla \vec{V}) - \frac{(1-\beta)^2}{(\beta^3 + \varepsilon)} A_{mush} \vec{V} + P_r \cdot \vec{n} + \rho \beta \vec{g} (T - T_m) \quad (2)$$

where P is pressure, μ is viscosity, ε is a small positive number to avoid division by zero, A_{mush} is a mushy zone constant. The third term on the right-hand side of Eq. (2) is the momentum sink due to the reduced porosity in the mushy zone. P_r is recoil pressure, $\vec{n}$ is the unit vector, β is thermal expansion coefficient, T_m is melting temperature.

Energy equation

$$\frac{\partial (\rho H)}{\partial t} + \nabla \cdot (\rho \vec{V} H) = \nabla \cdot (k \nabla T) + S_E \quad (3)$$

where H is the enthalpy, k is thermal conductivity, S_E is the energy source term.

The enthalpy-porosity technique [12,18] is adopted to deal with the solid-liquid phase transition. The enthalpy H is calculated as the sum of sensible heat h and the latent heat ΔH . The whole enthalpy is expressed by the formula

$$H = h + \Delta H \quad (4)$$

$$h = h_{ref} + \int_{T_{ref}}^T c_p dT \quad (5)$$

$$\Delta H = f_l H_l \quad (6)$$

where c_p is the specific heat capacity, T_{ref} and h_{ref} is the reference temperature and enthalpy at T_{ref} (300 K), H_l is the latent heat of melting. f_l is the liquid fraction and defined as the function of temperature

$$f_l = \begin{cases} 0 & T < T_s \\ \frac{T-T_s}{T_l-T_s} & T_s < T < T_l \\ 1 & T > T_l \end{cases} \quad (7)$$

where T_s and T_l are the solidus and liquidus temperature, respectively.

2.2. Laser heat source

In the paper, a moving adaptive hybrid heat source coupled with keyhole profile is used. As illustrated in Fig. 1, the hybrid heat source consists of a double ellipsoid heat source and a linear attenuation peak Gaussian heat source. The energy distribution changes in real time as the keyhole profile evolves, as shown in Fig. 2. Therefore, the hybrid heat source can be expressed by the following equation

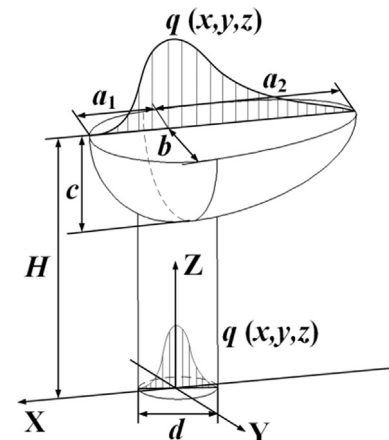


Fig. 1. Schematic of the hybrid heat source.

$$\begin{aligned}
q(x, y, z, t) &= \frac{6\sqrt{3}f_1k_1Q}{a_1bc\pi\sqrt{\pi}} \exp\left(-3\left(\frac{(x+L-vt)^2}{a_1^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\right)\right), H-c \leq z \leq H, x+L-vt > 0 \\
q(x, y, z, t) &= \frac{6\sqrt{3}f_1k_2Q}{a_2bc\pi\sqrt{\pi}} \exp\left(-3\left(\frac{(x+L-vt)^2}{a_2^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2}\right)\right), H-c \leq z \leq H, x+L-vt \leq 0 \\
q(x, y, z, t) &= \frac{6f_2Q}{\pi R^2 H(t)(1-e^{-3})} \frac{mz+R}{mH(t)+2R} \exp\left(-3\frac{(x+L-vt)^2+y^2}{R^2}\right), 0 \leq z < H-c
\end{aligned} \quad (8)$$

$$H(t) = H - Z(t - \Delta t) \quad (9)$$

where $q(x, y, z, t)$ is the power density at any point inside the source, Q is the laser power, H is the plate thickness (7 mm), $H(t)$ is the real-time keyhole depth, $Z(t-\Delta t)$ is the minimum ordinate value of the keyhole at the previous time step, L is the initial location of laser beam, f_1 and f_2 respectively represent proportion of two heat sources, $f_1 + f_2 = 1$. k_1 and k_2 represent proportion of two fractions of double-ellipsoid heat source respectively, $k_1 + k_2 = 2$. R is the radius of linear attenuation peak heat source, m is peak adjustment coefficient. a_1 , a_2 , b and c are parameters for the double-ellipsoid heat source.

In Eq. (8), R is set as 0.3 mm according to Xia et al. [19]. Parameters b and c for the double-ellipsoid heat source are determined by the practical weld bead profiles. a_1 and a_2 are adjusted based on the molten pool observed by high-speed imaging. Distribution coefficients f_1 and f_2 for the double-ellipsoid heat source and linear attenuation peak Gaussian heat source, adjustment coefficient m , are adjusted by numbers of simulation until the results accord well with experimental results.

$q(x, y, z, t)$ is the energy source term S_E in the Eq. (3).

2.3. Evaporation and condensation

Evaporation and condensation model considers energy transmission coupled with the mass transformation between liquid

phase and vapor phase. The governing equation of mass transforming mechanism is described as [20]

$$\frac{\partial(f_v \rho_v)}{\partial t} + \nabla \cdot (f_v \rho_v \vec{v}_v) = m_{lv} - m_{vl} \quad (10)$$

where f_v , ρ_v , $\vec{v}_v$ are respectively the volume fraction of vapor, vapor density and vapor velocity, while m_{lv} and m_{vl} are the mass transforming ratio from liquid to vapor and from vapor to liquid.

It's noted that the liquid phase transforms to vapor phase when the temperature of the computational cell exceeds the vaporization temperature, and the mass transforming ratio is calculated by

$$m_{lv} = \alpha_{lv} f_l \rho_l \frac{(T - T_v)}{T_v} \quad (11)$$

where α_{lv} , f_l , ρ_l respectively represents the evaporation coefficient, the liquid volume fraction, and the liquid density.

When the temperature of the computational cell reduces to below the vaporization temperature, the vapor phase transforms to liquid phase. The corresponding mass transforming ratio is calculated by

$$m_{vl} = \alpha_{vl} f_v \rho_v \frac{(T - T_v)}{T_v} \quad (12)$$

where α_{vl} , f_v , ρ_v respectively represents the transfer coefficient from vapor to liquid, vapor volume fraction, and the vapor density.

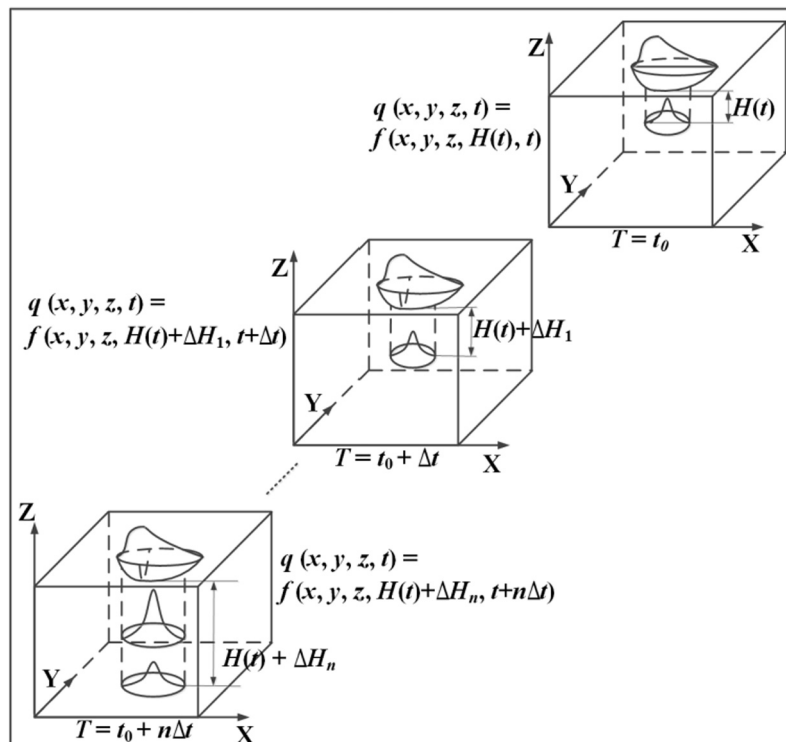


Fig. 2. The energy distribution of heat source.

m_{lv} and m_{vl} are the mass source terms in Eq. (1) for the liquid and vapor phases, respectively.

2.4. Tracking of keyhole surface

VOF method is used to track the free surface of the keyhole. Function $F(x, y, z, t)$ is defined to indicate the volume fraction of fluid in a cell and it satisfies the following equation

$$\frac{dF}{dt} = \frac{\partial F}{\partial t} + (\vec{V} \cdot \nabla)F = 0 \quad (13)$$

According to the definition, a cell with $F = 1$ is full of fluid, whereas $F = 0$ corresponds to a cell containing no fluid. A cell with an F value between 0 and 1 contains a free surface of the keyhole.

2.5. Boundary conditions

The analytic domain was set as 15 mm in length, 7 mm in width, and 12 mm in height including 7 mm plate thickness and 5 mm void regions at top for considering metallic vapor. Fig. 3 shows the sketch of half the domain in the simulation of high power laser welding process. Totally, 80640 uniform cells were used in the computation with the minimum cell is $0.25 \text{ mm} \times 0.25 \text{ mm}$.

The top surface of the computational domain is set as pressure outlet and ambient temperature. The upper domain is full of air in the beginning.

The side wall and bottom of the workpiece is treated as wall boundary condition which considers heat convection and radiation. The energy transfer can be described as

$$k \frac{\partial T}{\partial n} = -h_0(T - T_0) - \sigma \varepsilon(T^4 - T_0^4) \quad (14)$$

where h_0 and ε are heat convection coefficient and radiation coefficient, σ is Stefan-Boltzmann constant, T_0 is the ambient temperature.

On the keyhole free surface, energy of laser input, heat convection, heat radiation and liquid evaporation are considered. The energy balance equation is described as

$$k \frac{\partial T}{\partial n} = q - h_0(T - T_0) - \sigma \varepsilon(T^4 - T_0^4) - WH_v \quad (15)$$

where W is evaporation rate and H_v is the latent of evaporation.

Additionally, cells situated on the keyhole free surface are also subjected to pressures such as recoil pressure, surface tension, hydrostatic pressure and hydrodynamic pressure. In this paper, surface tension P_σ and recoil pressure P_r are mainly considered, as shown in Fig. 4.

The pressure equilibrium on the keyhole surface is as follow

$$P = P_\sigma + P_r \quad (16)$$

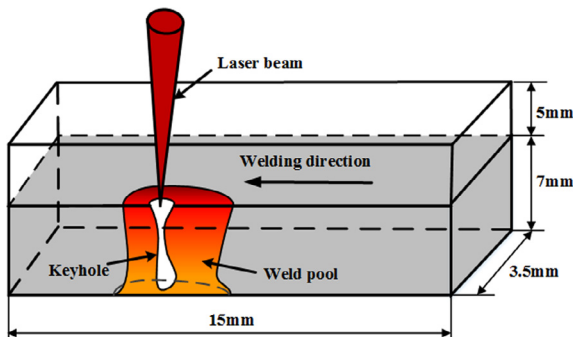


Fig. 3. Schematic of high power laser welding process.

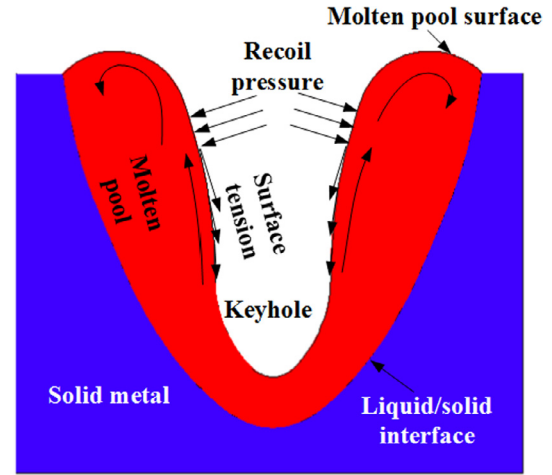


Fig. 4. Schematic of fluid flow and pressure equilibrium.

Surface tension P_σ is influenced by the surface curvature K and is calculated by

$$P_\sigma = K\gamma \quad (17)$$

where K is the curvature of free surface, γ represents the surface tension coefficient and expressed by [21]

$$\gamma = 1.943 - 4.3 \times 10^{-4}(T - 1710) - 1.3 \times 10^{-8}RT \times \ln \left[1 + 0.00318f_{\text{sulf}} \exp \left(\frac{1.66 \times 10^8}{RT} \right) \right] \quad (18)$$

where R is gas constant, f_{sulf} is sulfur concentration.

Recoil pressure on the keyhole free surface is calculated by [22]

$$P_r = 0.54P_0 \exp \left(H_v \frac{T - T_v}{RTT_v} \right) \quad (19)$$

where P_0 is the atmospheric pressure, H_v is the latent heat of evaporation, T_v represents the boiling temperature and R is the universal gas constant.

3. Experiment details

In this paper, a series of high power laser welding experiments of 7 mm thick 316L stainless steels were conducted. The laser system is a continuous wave fiber laser (IPG YLR-10000) with a maximum power of 10 kW. The spot diameter is 0.4 mm at the focal position. The composition of the base metal is listed in Table 1. Workpiece surfaces were cleaned with acetone in advance to avoid oil pollution. Through adjusting the welding parameters of laser power, welding speed and defocus amount, the experiment aimed to acquire different kinds of weld bead profiles. To study the fluid dynamics, high-speed CMOS camera was adopted to capture the instantaneous behavior of the keyhole and weld pool. After welding, all the samples were cut, ground, polished and etched. Then dimensions of the weld cross-sections were measured using the optical microscope.

Table 1
Material composition of 316L stainless steel.

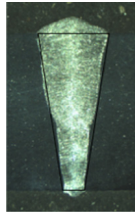
Element	Cr	Mn	Mo	Ni	Si	C	Fe
Content (wt.%)	17.3	1.74	2.66	13.1	0.73	0.023	Balance

4. Results and discussion

4.1. Experimental results

As the process parameters change, the weld bead profile and dimensions change accordingly. According to the cross sections of weld beads acquired under the optical microscope, the weld

Table 2
Three types of weld bead profiles.

Classification	Nail-shaped weld bead	Wedge-shaped weld bead	Slim-waist weld bead
Sample			
Process Parameters	Laser Power: 7k W Welding Speed: 30 mm/s Focus Position: 0 mm	Laser Power: 5.7 kW Welding Speed: 25 mm/s Focus Position: +20 mm	Laser Power: 8 kW Welding Speed: 20 mm/s Focus Position: -4 mm

bead profiles can be classified into three typical categories, i.e., the nail-shaped weld bead, the wedge-shaped weld bead and the slim-waist weld bead. As shown in Table 2, the three types of weld bead profiles present quite distinct features. Obviously, the nail-shaped weld bead has a larger width and steep slope in the upper zone, and in the middle and bottom zone, the profile resembles a slim and long columnar. The wedge-shaped weld bead presents the largest width in the upper zone but a linearly decreasing width along the depth. The slim-waist weld bead characterizes with a larger width in the upper and bottom zone but a smoothly decreasing width towards the middle zone which is relatively slimmer.

After analyzing the whole samples, the process parameters of different sorts of weld bead profiles followed certain regularities, as shown in Fig. 5. It seems that the slim-waist weld beads can be obtained when the line energy is more than 400 J/mm regardless of the power density. When the line energy is less than 250 J/mm, nail-shaped weld beads will occur. The wedge-shaped weld beads are in a transitional state between these two line energy with lower power density.

To get a better understanding of the influence of process parameters on the weld bead profile, it's essential to conduct simulation of the welding process, analyze the instantaneous behavior of the keyhole and weld pool, and find the main factors that determine the profile transformation.

4.2. Validation of the simulation model

Physical properties of the material and other parameters in the simulation are listed in Tables 3 and 4. The properties of molten pool were calculated by a commercial physical properties calculation software named JMatPro. The properties of metallic vapor and air were from the reference [7,12].

To ensure the rationality of the mathematical model, the simulated results of the three typical weld bead profiles were firstly compared with the experimental results. As shown in Fig. 6, the macroscopic profiles of the three weld beads containing both the experimental and simulation results are listed respectively. It is

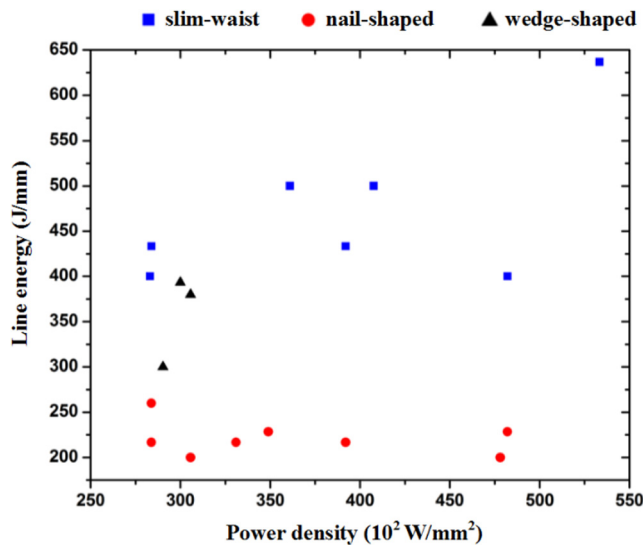


Fig. 5. Process parameters distribution of three types of weld beads.

Table 4
Other constants in the simulation.

Property	Symbol	Value
Sulfur concentration (ppm)	f_{sulf}	100
Heat convection coefficient ($W m^{-2} K^{-1}$)	h_0	30
Radiation coefficient	ε	0.4
Stefan-Boltzmann constant ($W m^{-2} K^{-4}$)	σ	5.67×10^8
Gas constant ($J mol^{-1} K^{-1}$)	R	8.314
Ambient temperature (K)	T_0	300

Table 3
Material physical properties.

Property	Symbol	Molten pool	Vapor	Air
Solidus/Liquidus/Boiling temperature (K)	$T_s/T_l/T_v$	1673/1710/3270	–	–
Density (kg/m^3)	ρ	$-0.51 T + 8053$, $300 K < T < 1673 K$ $-5.41 T + 16243$, $1673 K \leq T < 1710 K$ $-0.653 T + 8116$, $1710 K \leq T < 3270 K$	1	1.225
Viscosity ($kg/m s$)	μ	0.008, $1673 K < T < 1710 K$ $-1.919 \times 10^{-6} + 0.011$, $1710 K \leq T < 3270 K$	$3e-4$	1.789×10^{-5}
Thermal conductivity ($W m^{-1} K^{-1}$)	k	$0.0127 T + 11.798$, $300 K < T < 1673 K$ $-0.0811 T + 168.649$, $1673 K \leq T < 1710 K$ $0.0155 T + 3.415$, $1710 K \leq T < 3270 K$	15	0.0242
Specific heat capacity ($J/kg K$)	c_p	710, $300 K < T < 1673 K$ $3.243 T - 4715.54$, $1673 K \leq T < 1710 K$ 830, $1710 K \leq T < 3270 K$	371	1006.43
Latent heat of fusion/vaporization (J/kg)	H_f/H_v	2.77e5/6.24e6	–	–
Thermal expansion coefficient (K^{-1})	β	$2.6e-5$	–	–

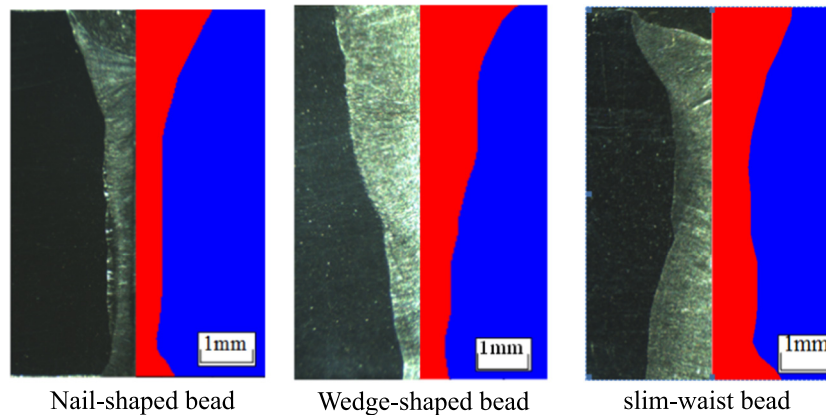


Fig. 6. Comparison of simulation and experimental results.

apparent that the predicted shapes coincide with the actual results. From the dimension sketch (Fig. 7) and the comparison chart (Fig. 8), it's undeniable that the dimensions of the predicted results match well with the actual values. Hence, the established model is quite rational to simulate the high power laser welding process.

4.3. Analysis of weld pool and keyhole behavior of the three types of weld beads

4.3.1. Analysis of weld pool and keyhole behavior of nail-shaped weld bead

The transient behavior of weld pool and keyhole under the condition of nail-shaped weld bead is shown in Fig. 9. From the profile of the keyhole and weld pool, it can be found that although the plate was fully penetrated, the keyhole still just penetrated.

The flow field presents three distinct vortices which are distributed in specific regions. The molten metal exhibits a large anticlockwise vortex at the molten pool bottom. From the figure, it can be seen that there exists small downward jetting vapor in the keyhole at the bottom, so the molten metal moves downwards with the vapor frictional force. And then it flows away from the molten pool center to the rear of the molten pool under the effect of Marangoni force. The fluid flow is prevented when it reaches to the solid/liquid interface, so the fluid begins to move upwards and forwards. Hence, a flow recirculation at the bottom of the weld pool is

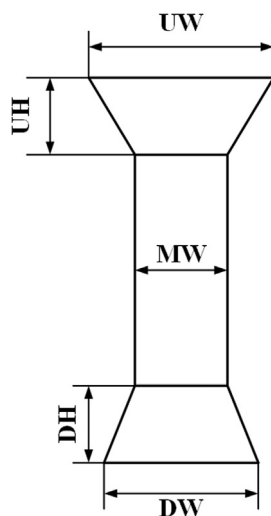


Fig. 7. The dimension sketch of weld bead profile.

formed. A small clockwise vortex can be found at the rear of the middle part inside the weld pool. From Fig. 9, it can be seen that the keyhole is much wider in the middle, and the vapor is much stronger. The big extrusion force of the vapor on the keyhole interface makes the fluid flows upwards and backwards. When the flow meets the solid/liquid interface, it is prevented and then flows downwards. In addition, the formation of the anticlockwise vortex at the upper part of the weld pool can be explained as below. From Fig. 9, it can be seen that on the upper part, the vapor extrusion force makes the flow has a big velocity near the rear keyhole interface. The fluid flows upwards and backwards, and when it reaches the solid/liquid interface, it is impeded and then tends to backfills the keyhole vacancy. In addition, the vapor drag force and Marangoni force makes the fluid flows backwards at the top surface of the molten pool. Thus, the vortex is a combined result of these effects.

Fig. 10 shows the fluid flow of four typical positions AA₁, BB₁, CC₁ and DD₁ in Fig. 9. Fig. 10(a) represents the flow field of the keyhole center, from which it can be seen that the keyhole just penetrates the plate and is filled with high-temperature upward jetting vapor. Fig. 10(b) shows the flow field near the rear keyhole interface. A distinct vacancy is observed at this position. The region near the vacancy is filled with molten metal and the fluid flows around the vacancy. The existing vacancy indicates that the recoil pressure here is inadequate to balance with the surface tension. However, in this case, porosity defects will occur more easily. Fig. 10(c) displays the flow field of the molten pool. The fluid can be divided into two regions. The upper half part in the molten pool flows upwards and the velocity becomes larger with the height, while the lower half part in the molten pool flows downwards and the velocity becomes smaller with the height. In the middle, the velocity almost reduces to zero. If the flow field is associated with the temperature field, it can be found that the temperature at the lower half part is relatively much smaller and the viscosity is correspondingly larger which results in slight movement at the bottom. Fig. 10(d) illustrates the flow field of the solid/liquid interface. It is found that the solidification rate is not the same along the depth. While a part of the fluid has already solidified, some of the fluid is still in the mushy status. As mentioned above, the high-temperature fluid at the bottom flows backwards due to the Marangoni force and accumulates near the solid/liquid interface, the rear part of the bottom is reheated. Hence the solidification at the bottom is delayed and the molten pool at the bottom is wider than the middle part.

4.3.2. Analysis of weld pool and keyhole behavior of wedge-shaped weld bead

The instantaneous behavior of weld pool and keyhole under the condition of wedge-shaped bead is depicted in Fig. 11.

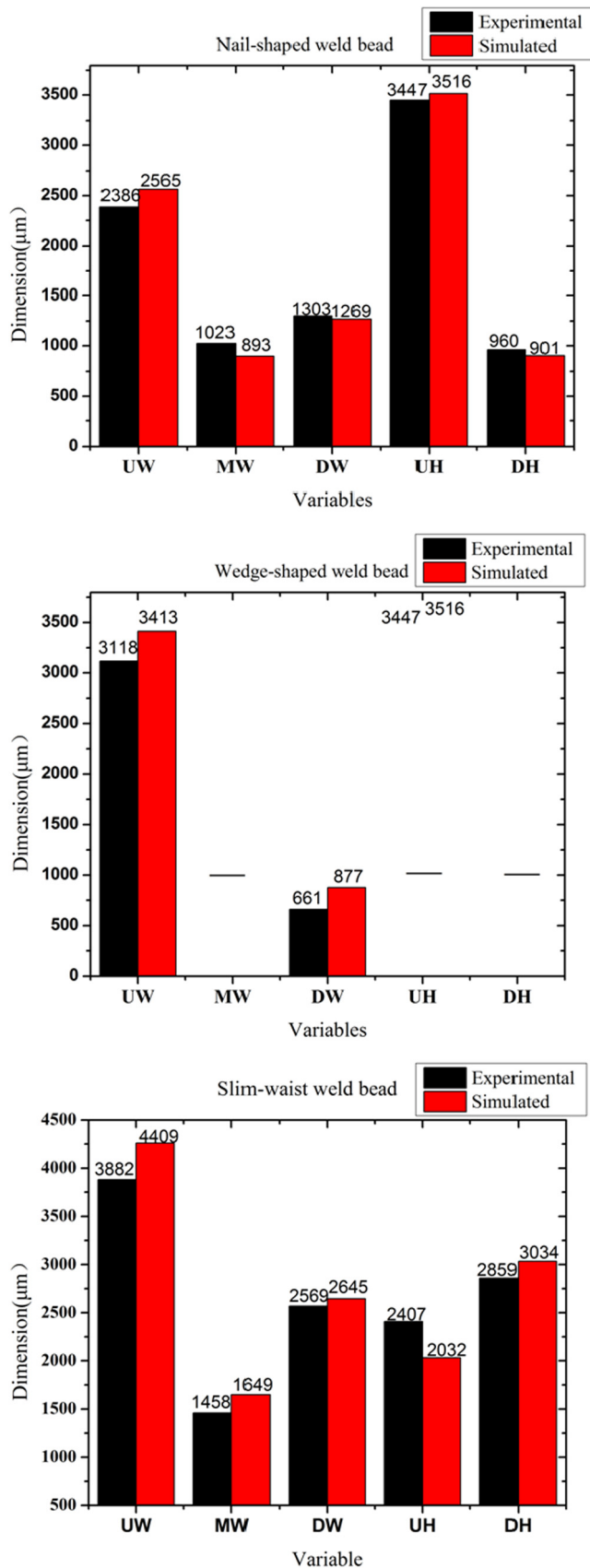


Fig. 8. Dimension comparison of the three types of weld bead profiles.

Compared with Fig. 9, it's apparent that this case is quite distinct from that of nail-shaped bead whether in flow field or the profile of molten pool and metal vapor. For the laser power was only 5.7 kW and the defocus amount was up to +20 mm, the keyhole couldn't penetrate the steel plate even the welding speed was slower than the nail-shaped case. The reason for the anticlockwise vortex at the bottom of the molten pool is due to the Marangoni force. A big clockwise vortex distributes in the whole molten pool. It can be explained as follows: Fluid flows upwards with the effect of upward jetting vapor and then flows away from the molten pool center to the solid/liquid interface because of the Marangoni force. Fluid is prevented by the solid/liquid interface and flows downwards along the interface, and thus a big clockwise vortex forms. Another small clockwise vortex appears in the middle of the molten pool. From the Fig. 11, it can be seen that the keyhole is wider in the middle. The bigger extrusion force combined with the upward vapor frictional force makes a small clockwise vortex.

Fig. 12 displays the flow field of the AA1 cross section in Fig. 11. From the figure it can be seen that, the width of molten pool becomes smaller along the depth and the width of the molten pool's upper surface is much larger than that of the nail-shaped one. As for the reason, it could be explained as below. Firstly, the defocus amount reached +20 mm, it indicates that the surface area irradiated by the laser beam is larger and the power density decreases sharply along the depth, so this will definitely contribute to widening the molten pool at the surface and smaller width along the depth. Secondly, from the flow field it seems that the keyhole keeps wide, which means the vapor recoil pressure in the keyhole would result in continuous extrusion force to push away the fluid adjacent to the keyhole interface. Thirdly, the large amount of high-temperature metal vapor at the keyhole opening contributes to great temperature gradient, thus the Marangoni convection will be stronger. For the surface tension coefficient is negative, the surface tension becomes larger as the temperature reduces, the fluid along the molten pool surface will flow away from the keyhole interface towards the low-temperature region. From all above reasons, the wedge-shaped weld bead appears decreasing width along the depth and a larger width at the upper surface than that of the nail-shaped one.

4.3.3. Analysis of weld pool and keyhole of slim-waist weld bead

The keyhole and molten pool dynamics of slim-waist weld bead also has its own characteristics. Fig. 13 exhibits the flow field of the longitudinal section. The most significant feature in this case is the fully penetrated keyhole. Both the weld bead profile and the flow field are obviously different from that of the other two cases. The anticlockwise vortex at the bottom is due to the drag force of the vapor and the Marangoni force. The fluid at the rear of the molten pool flows upwards along the solid/liquid interface until to the upper surface of the molten pool. Additionally, the fluid adjacent to the keyhole interface also flows upwards to the upper surface, along the keyhole interface. It's not difficult to explain this phenomenon. In this case, the laser power is 8 kW and the line energy reaches 400 J/mm. The large energy makes the full penetration welding. From the figure, it can be seen that there exists much downward jetting vapor. So, liquid flows towards the rear of the molten pool bottom under the effect of downward jetting vapor and Marangoni force. The fluid flow is prevented when they get to the solid/liquid interface. Meanwhile, high speed upward jetting vapor exerts a large upward drag force on the molten pool. So, the blocked liquid flows upwards to the upper surface with the high

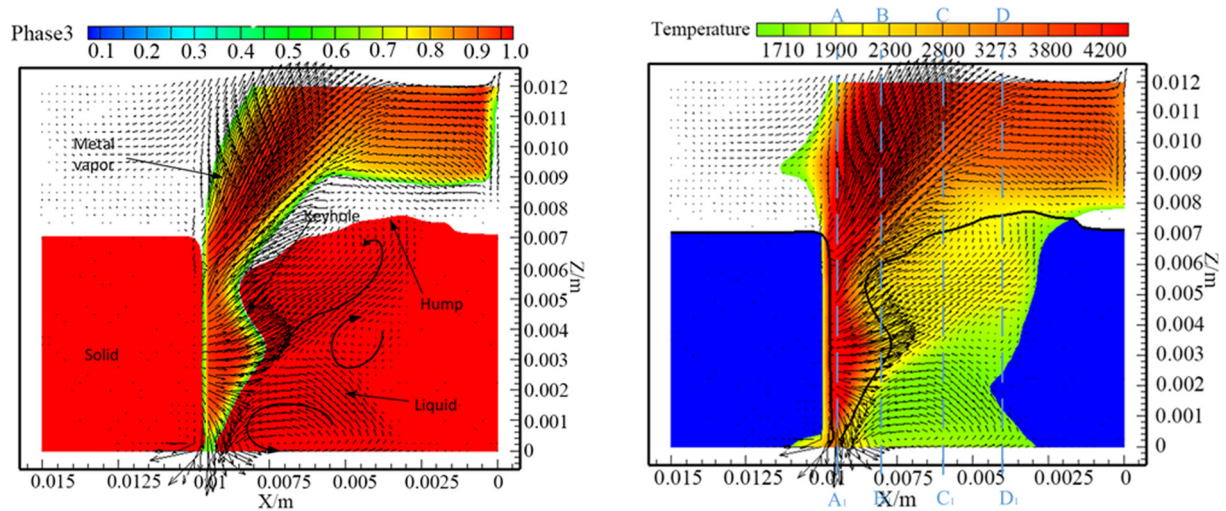


Fig. 9. Flow field of the longitudinal section at $t = 0.22$ s of nail-shaped case.

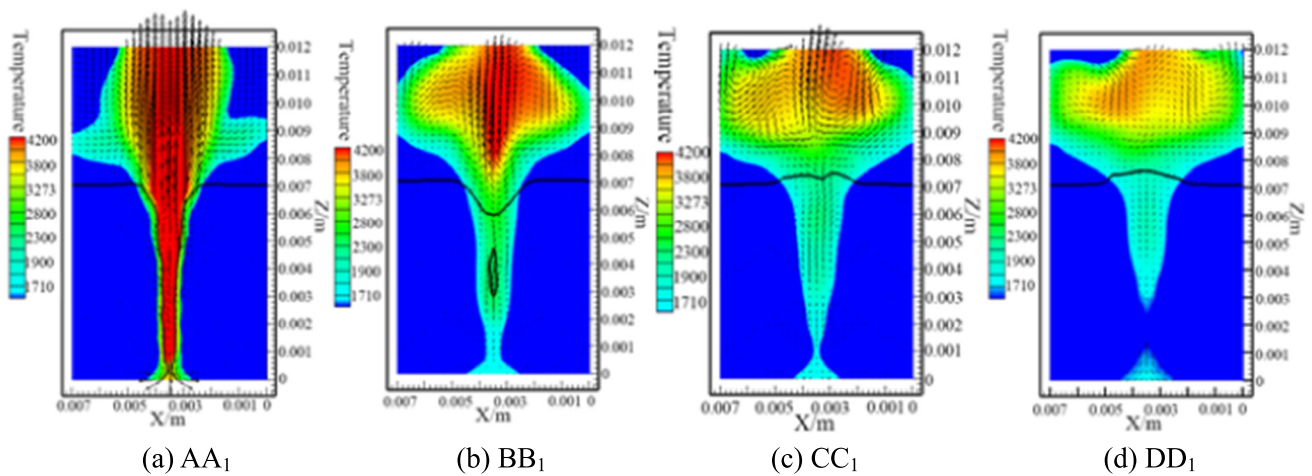


Fig. 10. Flow field of the AA₁, BB₁, CC₁ and DD₁ cross section in Fig. 9.

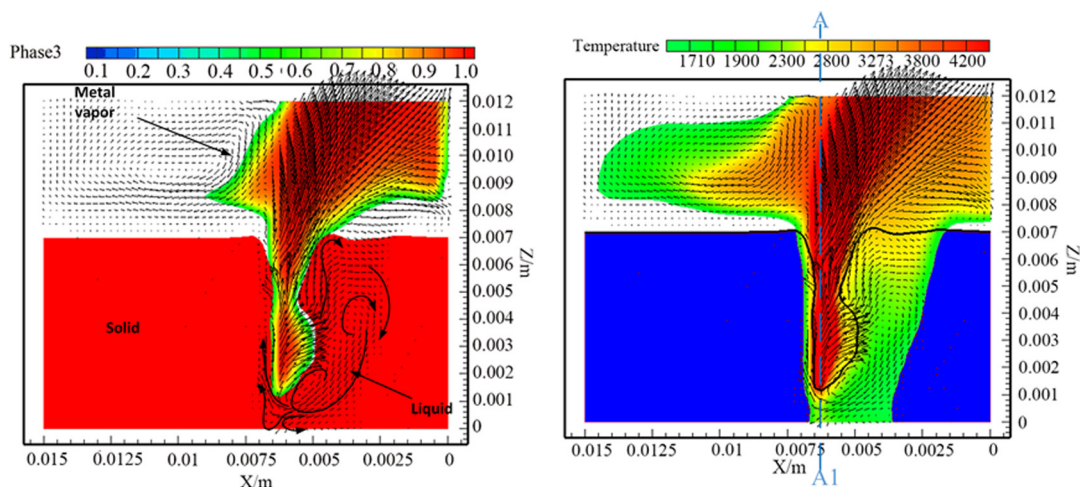


Fig. 11. Flow field of the longitudinal section at $t = 0.22$ s in wedge-shaped case.

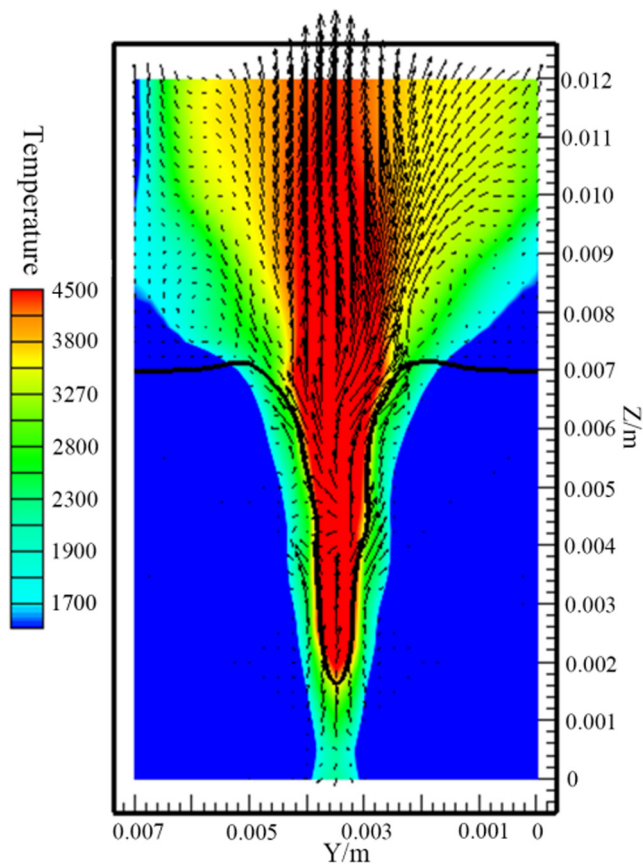


Fig. 12. Flow field of the cross section's fluid flow of AA1 in Fig. 11.

speed upward vapor. Then with the effect of Marangoni force at the top surface, they flow to the rear of the molten pool.

Compared with the flow field of cross section of the nail-shaped weld bead, it is similar but still has several special characteristics.

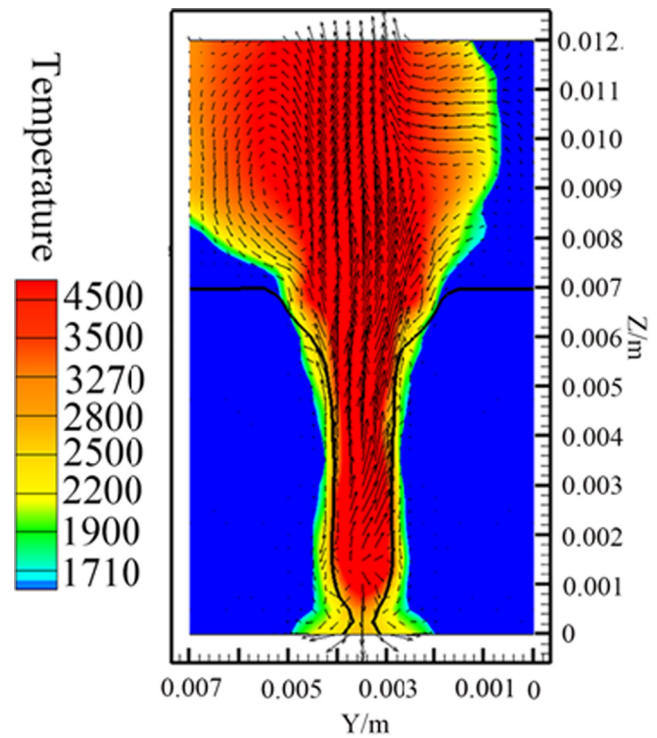


Fig. 14. Flow field of the cross section's fluid flow of AA1 in Fig. 13.

As shown in Fig. 13, the keyhole has penetrated the plate thoroughly, so more laser energy can transfer to the bottom of the keyhole compared with the nail-shaped case. Thus, the width of the slim-waist weld bead is larger in the lower part. In addition, liquid flows towards the rear of the molten pool bottom under bigger vapor drag force and Marangoni force, so the bottom of the molten pool is relatively much wider than that of the nail-shaped weld bead. All these can be seen from Figs. 14 and 10.

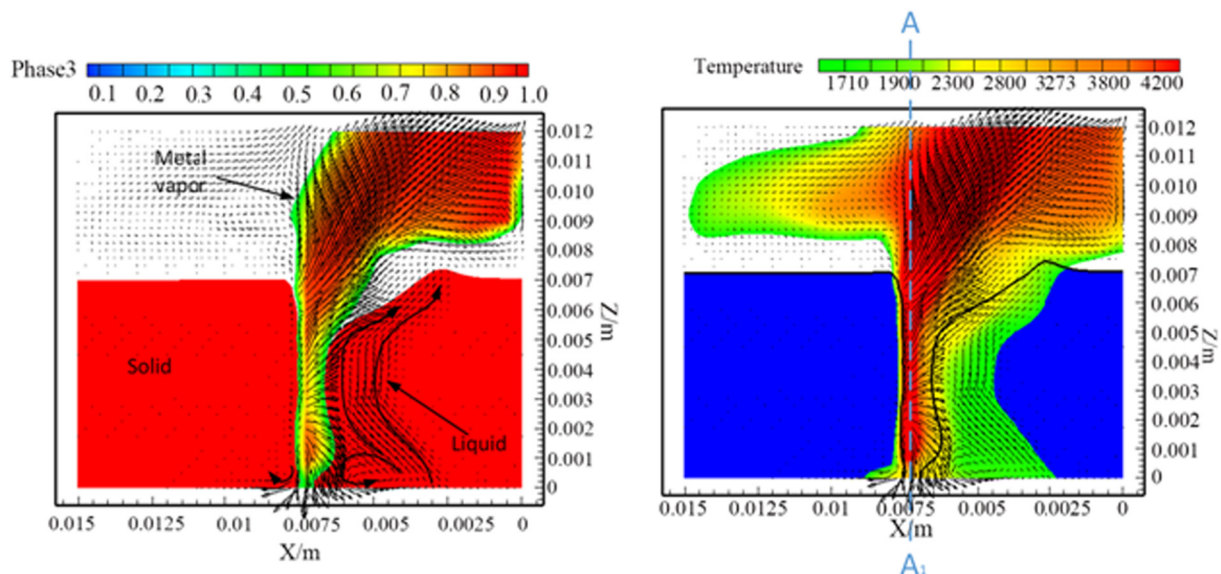


Fig. 13. Flow field of the longitudinal section at $t = 0.22$ s in slim-waist case.

5. Conclusion

A three-dimensional transient model has been developed to study the weld bead profiles transforming mechanism in the high power laser welding process. The following conclusions can be obtained:

- (1) The weld bead profiles are mainly determined by line energy. Slim-waist weld beads can be obtained when the line energy is more than 400 J/mm regardless of the power density. When the line energy is less than 250 J/mm, nail-shaped weld beads will occur. Wedge-shaped weld beads are in a transitional state between these two line energy with lower power density.
- (2) Whether the keyhole penetrates the plate is the main factor that affects the bottom of weld bead profiles. The bottom weld bead profile tends to be wider when the keyhole penetrates the plate thoroughly, otherwise the bottom weld bead profile will be narrower.
- (3) Fluid flow of the three types of weld bead profiles is different. The recoil pressure acting on the keyhole interface, the Marangoni force derived from the temperature gradient and the frictional force derived from the ejection of the metal vapor influence the flow field of the molten pool, and these momentum factors together with the temperature field simultaneously determine the whole weld bead profile.

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Conflict of interest

We declare that we have no financial and personal relationships with other people or organizations that can inappropriately influence our work, there is no professional or other personal interest of any nature or kind in any product, service and/or company that could be construed as influencing the position presented in, or the review of, the manuscript entitled "Investigation on the weld bead profile transformation with the keyhole and molten pool dynamic behavior simulation in high power laser welding".

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